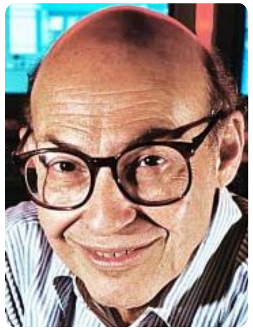
prof. dr Aleksandar Milosavljević

Računarski vid
Computer Vision (CV)

Istorijat
Čulo vida




Kratak istorijat računarskog vida

- Začeci se vezuju za rane 1970-te i veštačku inteligenciju
- Računarski vid jedna komponenta za ostvarivanje VI i razvoja robota
- Pioniri VI su smatrali da je rešavanje „vizuelnog ulaza“ mali korak na putu rešavanja složenih problema kao što su rezonovanje i planiranje
- Postoji anegdota da je 1966. god. Marvin Minski sa MIT-a dao zadatak svom studentu Džeraldu Džeju Sasmanu da za letnji projekat reši problem povezivanja kamere na računar tako da računar bude u stanju da opiše šta je video
  - Problem nije rešen za jedno leto, ali je zato postavka problema i dalje aktuelna


MACHINE PERCEPTION OF THREE-DIMENSIONAL SOLIDS

by

LAWRENCE GILMAN ROBERTS

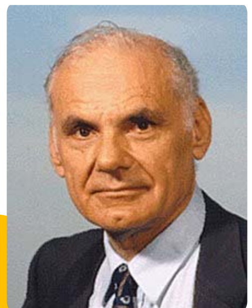
S.B., Massachusetts Institute of Technology (1961)

M.S., Massachusetts Institute of Technology (1961)

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

at the

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
June, 1963



Kratak istorijat računarskog vida

- Ono što je izdvojilo računarski vid od do tada postojećeg polja obrade slika (eng. *image processing*) je bio cilj da se iz slika izvuče trodimenzionalna (3D) struktura scene, te da se to iskoristi za razumevanje scene
- Prvi korak ka ovom cilju je učinjen od strane Larija Robertsa koji je u svojoj doktorskoj disertaciji iz 1963. godine, odbranjenoj na MIT-ju, diskutovao mogućnosti za ekstrakciju 3D geometrijskih informacija iz 2D slika blokova
- Mnogi istraživači iz ovog perioda su nastavili da slede njegov primer, tako da su prva istraživanja u računarskom vidu i veštačkoj inteligenciji bila ograničena na svet blokova

Kratak istorijat računarskog vida

1970-te
- Detekcija ivica i linija
- Ne-poliedarsko i poliedarsko modelovanje
- Predstavljanje objekata na osnovu povezanih delova
- Optički tok i procena kretanja

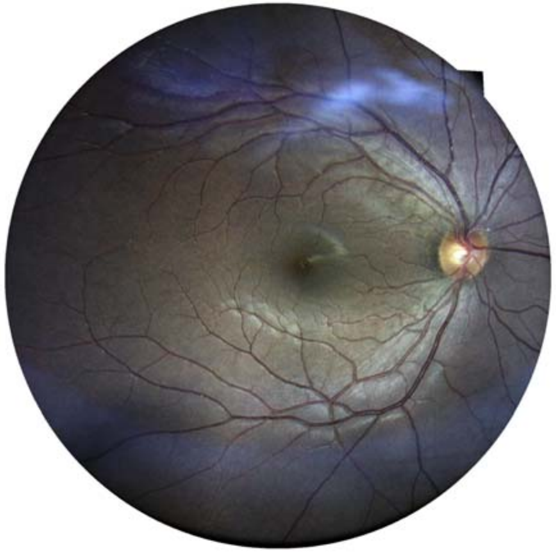
1980-te
- Piramide slika (*image blending*, gruba-ka-finoj korespodencija)
- Procena oblika na osnovu senčenja (*shape from shading*)

1990-te
- 3D rekonstrukcija i kalibracija kamere
- Segmentacija slika (*graph cut*)
- *Eigenfaces*
- *Image morphing*, interpolacija pogleda, spajanje slika u panoramu

2000-te
- lokalna obeležja (*features*)
- prepoznavanje objekata
- mašinsko učenje

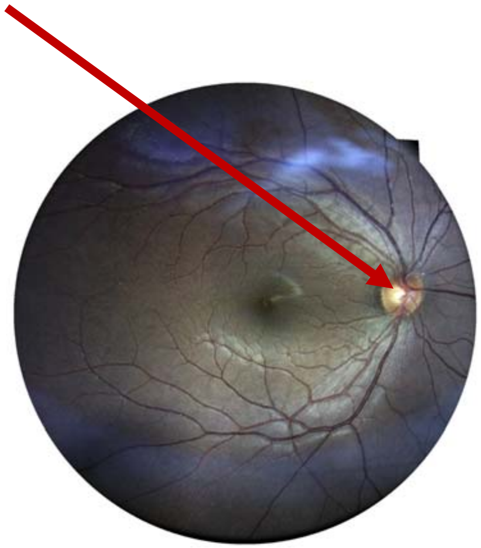
2010-te
- duboko učenje
- konvolucione mreže

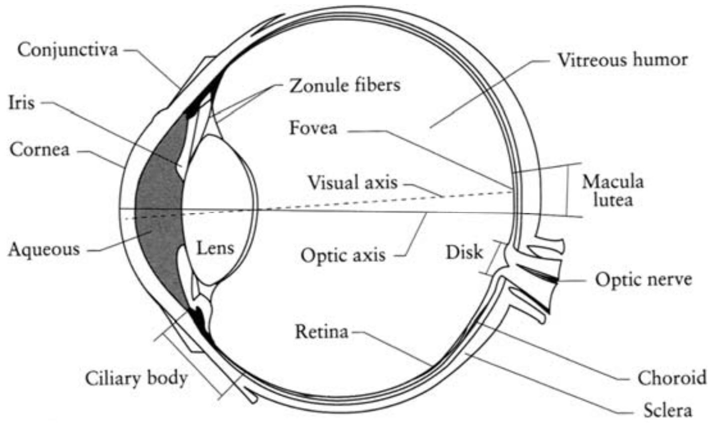
Čulo vida kod ljudi



Mrtva tačka (blind spot)

Mrtva tačka (blind spot)





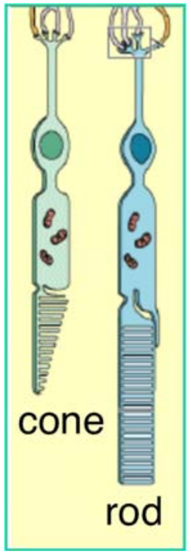
Ljudsko oko

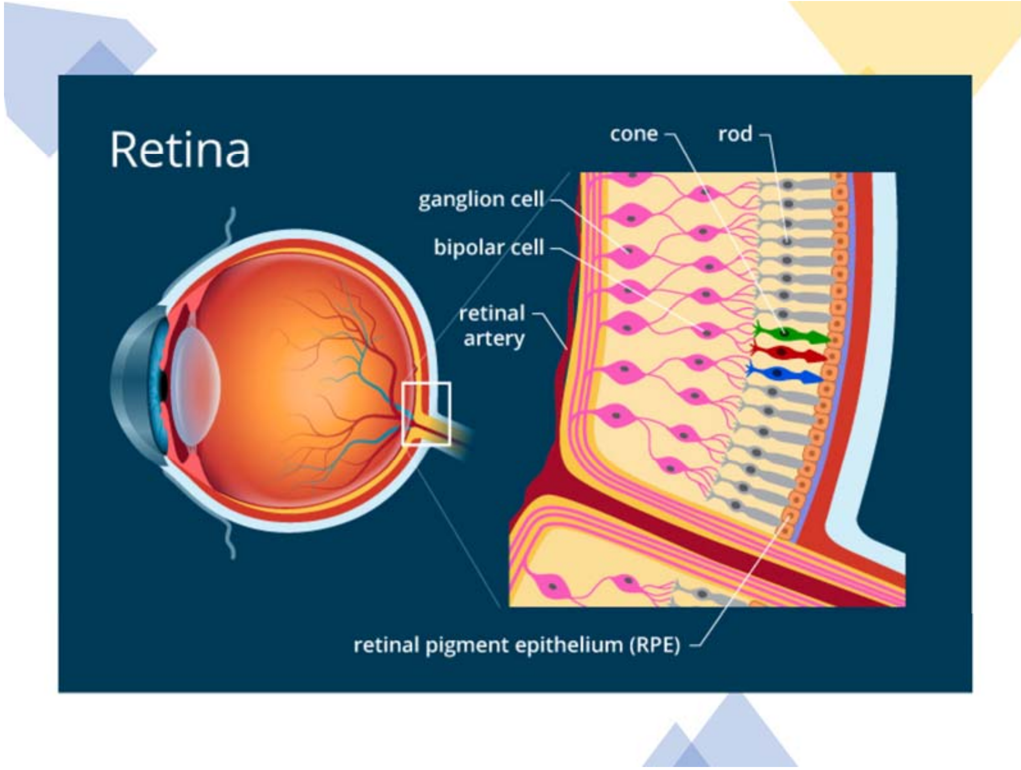
- Oko je analogno foto aparatu, tj. kameri
  - **Zenica (*pupil*)** – odgovara blendi kod foto aparata
  - **Iris** – obojeni prsten sa radijalnim mišićima, reguliše otvor „blende“ (aperturu)
- Šta odgovara “filmu”?
  - **Mrežnjača** (***retina***) koja sadrži fotoreceptorske ćelije:
    - **Štapići** (***rods***) – reaguju na osvetljaj
    - **Čepići** (***cones***) – reaguju na boju

Dva tipa receptora koji reaguju na svetlost

Čepići (cones)
- konusnog oblika
- manje osetljivi na svetlost
- aktiviraju se kada je jako svetlo
- zaduženi za percepciju boje

Štapići (rods)
- u obliku štapa
- visoko osetljivi na svetlost
- obezbeđuju monohromatski vid noću





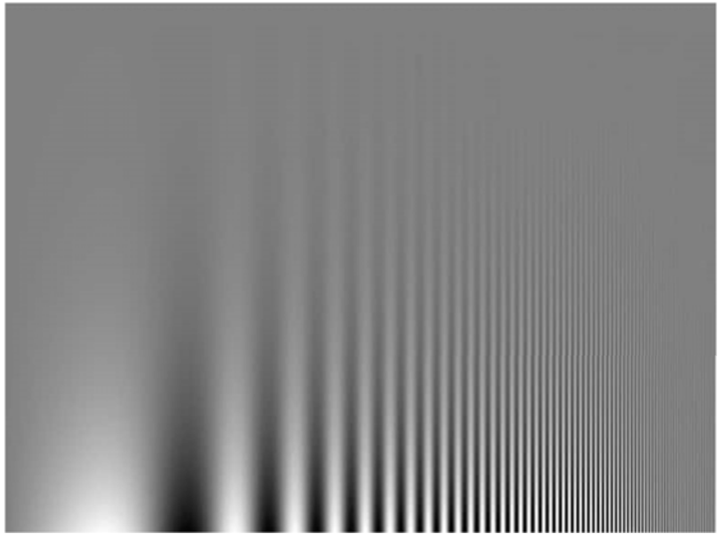
Periferni vid

- Pokušajte da izbrojite prste na ruci kada ne gledate direktno u nju
- Štapici (rods) 1000x osetljiviji na svetlo od čepića (cones)
- Raspodela čepića po bojama:
  - 64% crvena, 32% zelena, 4% plava
- Čepići mnogo veći i po periferiji, štapići u predelu žute mrlje (fovea)
- *Why we have blind spots - and how to see the blood vessels inside your own eye!*
  - https://www.youtube.com/watch?v=L_W-IXqoxHA

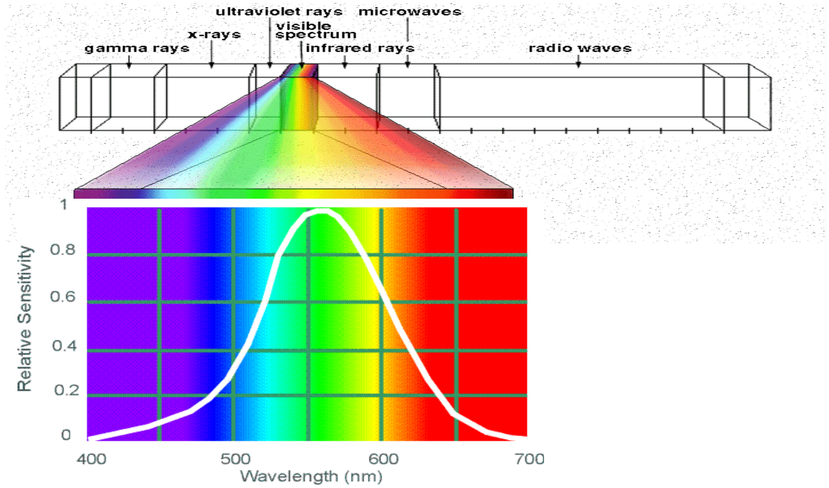
Obrada u mrežnjači (retina)

- HDR (High Dynamic Range) – vidimo jasno i unutra i spolja
  - Da bi se dobila ovakva slika kamerom potrebno je blendovanje više slika snimljenih pri različitim ekspozicijama
- Ganglioni zaduženi za obradu na najnižem nivou
  - Porede lokalno osvetljenje sa osvetljenjem u široj oblasti
  - Ovo proizvodi osetljivost na kontrast pri različitim frekvencijama (naredni slajd)

Osetljivost na kontrast

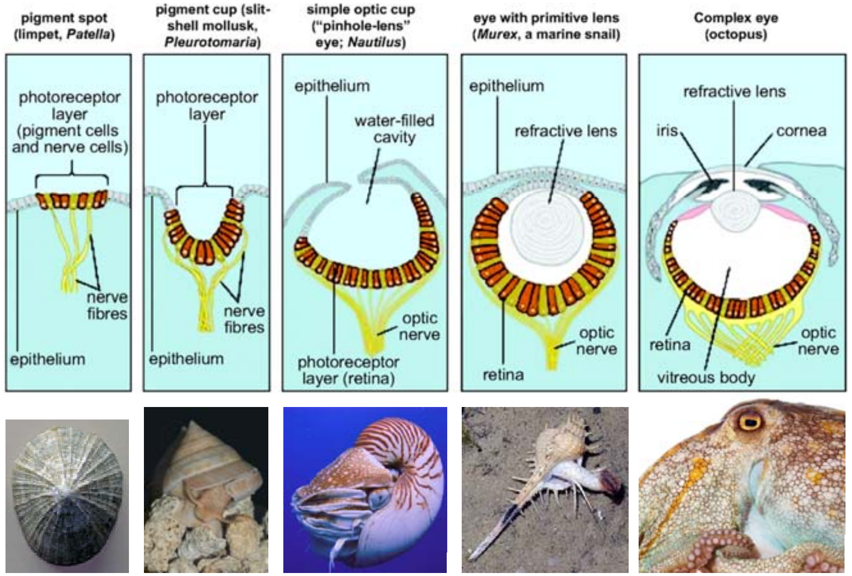


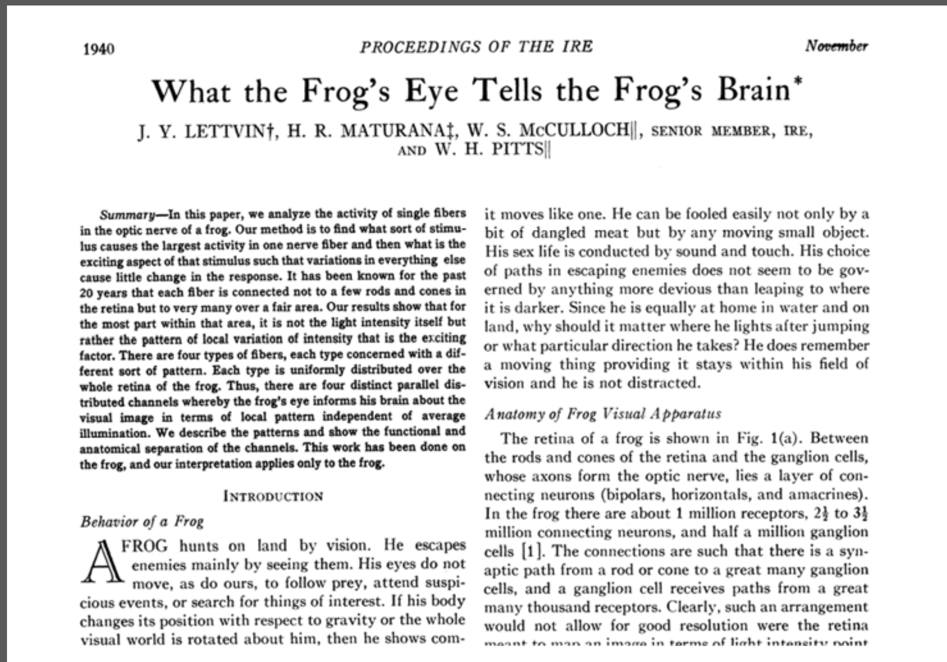
Elektromagnetski spektar



Osetljivost ljudskog vida na boje iz spektra

Evolucija oka kod mekušaca





1940 *PROCEEDINGS OF THE IRE* *November*

What the Frog's Eye Tells the Frog's Brain*

J. Y. LETTVIN†, H. R. MATURANA‡, W. S. McCULLOCH||, SENIOR MEMBER, IRE, AND W. H. PITTS||

Summary—In this paper, we analyze the activity of single fibers in the optic nerve of a frog. Our method is to find what sort of stimulus causes the largest activity in one nerve fiber and then what is the exciting aspect of that stimulus such that variations in everything else cause little change in the response. It has been known for the past 20 years that each fiber is connected not to a few rods and cones in the retina but to very many over a fair area. Our results show that for the most part within that area, it is not the light intensity itself but rather the pattern of local variation of intensity that is the exciting factor. There are four types of fibers, each type concerned with a different sort of pattern. Each type is uniformly distributed over the whole retina of the frog. Thus, there are four distinct parallel distributed channels whereby the frog's eye informs his brain about the visual image in terms of local pattern independent of average illumination. We describe the patterns and show the functional and anatomical separation of the channels. This work has been done on the frog, and our interpretation applies only to the frog.

INTRODUCTION

Behavior of a Frog

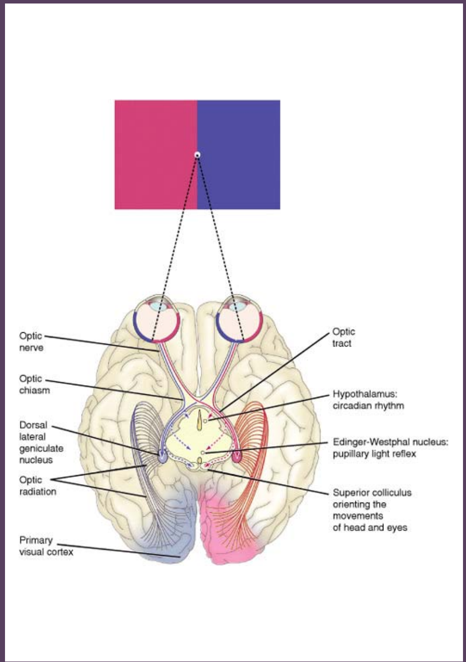
A FROG hunts on land by vision. He escapes enemies mainly by seeing them. His eyes do not move, as do ours, to follow prey, attend suspicious events, or search for things of interest. If his body changes its position with respect to gravity or the whole visual world is rotated about him, then he shows com-

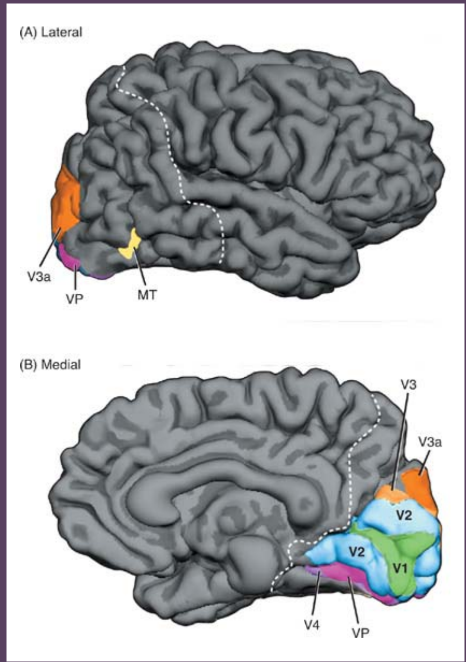
it moves like one. He can be fooled easily not only by a bit of dangled meat but by any moving small object. His sex life is conducted by sound and touch. His choice of paths in escaping enemies does not seem to be governed by anything more devious than leaping to where it is darker. Since he is equally at home in water and on land, why should it matter where he lights after jumping or what particular direction he takes? He does remember a moving thing providing it stays within his field of vision and he is not distracted.

Anatomy of Frog Visual Apparatus

The retina of a frog is shown in Fig. 1(a). Between the rods and cones of the retina and the ganglion cells, whose axons form the optic nerve, lies a layer of connecting neurons (bipolars, horizontals, and amacrines). In the frog there are about 1 million receptors, $2\frac{1}{2}$ to $3\frac{1}{2}$ million connecting neurons, and half a million ganglion cells [1]. The connections are such that there is a synaptic path from a rod or cone to a great many ganglion cells, and a ganglion cell receives paths from a great many thousand receptors. Clearly, such an arrangement would not allow for good resolution were the retina

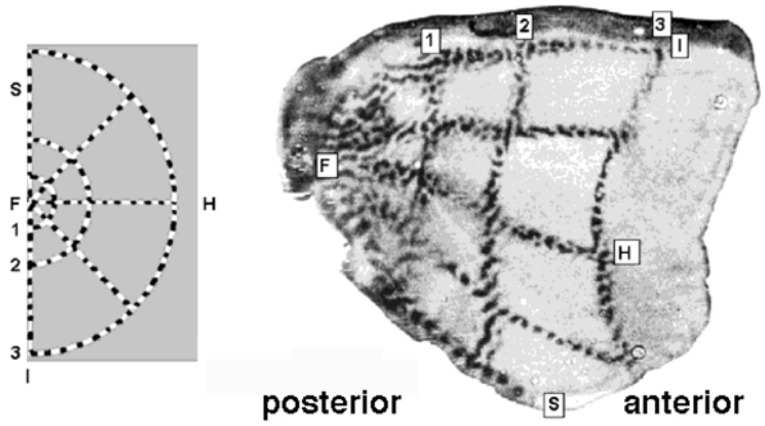
1959. god.





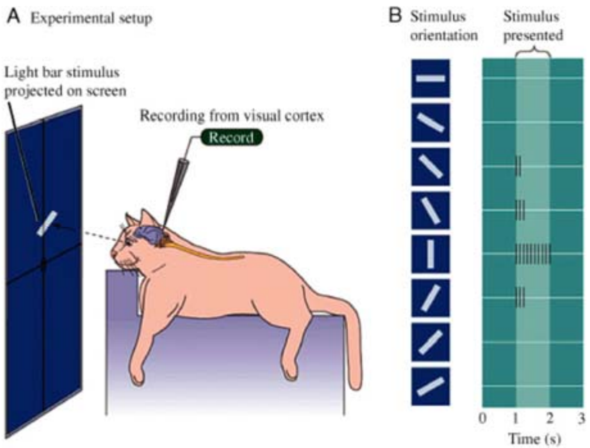
Vizuelni korteks

- Projektuje se slika vizuelnog stimulusa
- Eksperimenti sa životinjama



Eksperimenti Hubela i Vizela (Hubel & Wiesel)

Dobili Nobelovu nagradu 1981. za otkrića vezana za obradu vizuelnih informacija u mozgu



https://www.youtube.com/watch?v=IOHayh06LJ4

Obrada u korteksu

- V1 (primarni vizuelni korteks)
  - Zadužen za detekciju jednostavnih elemenata
  - Postoje ćelije koje reaguju na orijentaciju nadržaja (linije)
  - Postoje ćelije koje reaguju na pravac pomeranja
- V2 (sekundarni vizuelni korteks)
  - Ćelije reaguju na složenije strukture (npr. na ruku ili lica)
- Postoje delovi korteksa zaduženi za različite stvari
  - Kod povreda osoba može da izgubi određeno svojstvo vida (npr. motion blindness)
- Tok obrade vizuelnih nadražaja ide od „nižeg“ ka „višem“ nivou
  - Postoje i poprečne veze, nije sve tako jednostavno